```python
# 1. using map() to find squares of numbers
numbers = [1, 2, 3, 4, 5]

squares = list(map(lambda x: x * x, numbers))

print("Original List:", numbers)
print("Squares List:", squares)
```

[47] 119ms

```
Original List: [1, 2, 3, 4, 5]
Squares List: [1, 4, 9, 16, 25]
```

```python
# 2. using filter() to find even numbers
numbers = [10, 15, 20, 25, 30, 35, 40]

even_numbers = list(filter(lambda x: x % 2 == 0, numbers))

print("Original List:", numbers)
print("Even Numbers:", even_numbers)
```

[48] 106ms

```
Original List: [10, 15, 20, 25, 30, 35, 40]
Even Numbers: [10, 20, 30, 40]
```

```python
# 3. using reduce() to find sum of elements
from functools import reduce

numbers = [1, 2, 3, 4, 5]

sum_result = reduce(lambda x, y: x + y, numbers)

print("List Elements:", numbers)
print("Sum of Elements:", sum_result)

```

✓ [49] 51ms

```
List Elements: [1, 2, 3, 4, 5]
Sum of Elements: 15
```

```python
# 4. function returning another function
def multiplier(n):
    return lambda x: x * n

double = multiplier(2)
triple = multiplier(3)

print("Double of 5:", double(5))
print("Triple of 5:", triple(5))

```

✓ [50] 46ms

```
Double of 5: 10
Triple of 5: 15
```

```python
# 5. using sorted() with lambda function
students = [
    ("Anu", 85),
    ("Rahul", 72),
    ("Priya", 90),
    ("Kiran", 78)
]

sorted_students = sorted(students, key=lambda x: x[1])

print("Students Sorted by Marks:")
for student in sorted_students:
    print(student)
```

✓ [51] 39ms

```
Students Sorted by Marks:
('Rahul', 72)
('Kiran', 78)
('Anu', 85)
('Priya', 90)
```

```python
# 6. using map() to convert strings to uppercase
names = ["anu", "rahul", "priya", "kiran"]

uppercase_names = list(map(lambda x: x.upper(), names))

print("Original Names:", names)
print("Uppercase Names:", uppercase_names)
```

✓ [52] 40ms

```
Original Names: ['anu', 'rahul', 'priya', 'kiran']
Uppercase Names: ['ANU', 'RAHUL', 'PRIYA', 'KIRAN']
```

```python
# 7. using filter() to find positive numbers
numbers = [-10, 20, -30, 40, 50, -60]

positive_numbers = list(filter(lambda x: x > 0, numbers))

print("Original List:", numbers)
print("Positive Numbers:", positive_numbers)
```

✓ [53] 36ms

```
Original List: [-10, 20, -30, 40, 50, -60]
Positive Numbers: [20, 40, 50]
```

```python
# 8. using reduce() to find maximum number
from functools import reduce

numbers = [12, 45, 67, 23, 89, 34]

maximum = reduce(lambda x, y: x if x > y else y, numbers)

print("List Elements:", numbers)
print("Maximum Element:", maximum)
```

✓ [54] 53ms

```
List Elements: [12, 45, 67, 23, 89, 34]
Maximum Element: 89
```

```python
# 9. lambda function for addition
addition = lambda a, b: a + b

num1 = 10
num2 = 20

result = addition(num1, num2)

print("First Number:", num1)
print("Second Number:", num2)
print("Sum:", result)
```

✓ [55] 151ms

```
First Number: 10
Second Number: 20
Sum: 30
```

```python
# 10 using map() to calculate length of strings
words = ["Python", "Java", "C", "JavaScript"]

lengths = list(map(lambda x: len(x), words))

print("Words:", words)
print("Lengths:", lengths)
```

✓ [56] 30ms

```
Words: ['Python', 'Java', 'C', 'JavaScript']
Lengths: [6, 4, 1, 10]
```

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